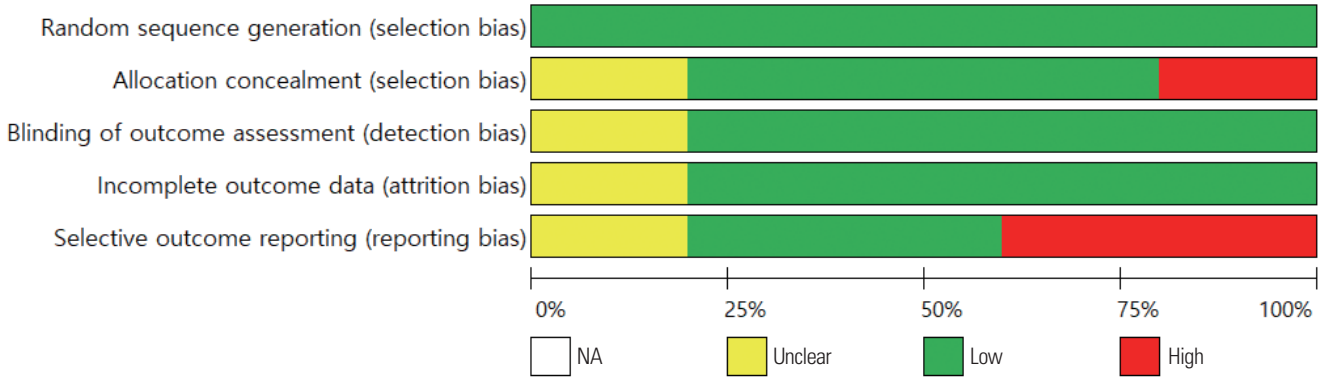
Supplementary Table 1. Search strategy for included studies

Data source		Search term
PubMed	#1	nondiabet*[TW] OR non-diabet*[TW] OR without-diabet*[TW]
	#2	normal*[TW] AND ("glucose"[MeSH] OR glucose*[TW])
	#3	"Prediabetic State"[Mesh] OR Prediabet*[TW] OR Pre-diabet*[TW]
	#4	#1 OR #2 OR #3
	#5	"Obesity"[Mesh]
	#6	"Overweight"[Mesh:noexp]
	#7	"Overnutrition"[Mesh:Noexp]
	#8	Obesit*[TW] OR Obese[TW] OR Overweight*[TW] OR Overnutrition*[TW] OR over-weight*[TW]
	#9	#5 OR #6 OR #7 OR #8
	#10	"Sodium-Glucose Transporter 2"[Mesh] OR "Sodium Glucose"[TW] OR "SGLT2"[TW] OR gliflozin
	#11	dapagliflozin[TW] OR "2-(3-(4-ethoxybenzyl)-4-chlorophenyl)-6-hydroxymethyltetrahydro-2H-pyran-3,4,5-triol" [Supplementary Concept] OR "BMS-512148"[TW] OR "BMS 512148"[TW] OR "BMS512148"[TW]
	#12	"empagliflozin"[Supplementary Concept] OR "empagliflozin"[TW] OR "BI-10773"[TW] OR "BI 10773"[TW] OR "BI10773"[TW]
	#13	"ipragliflozin"[Supplementary Concept] OR "ipragliflozin"[TW] OR "Suglat"[TW] OR "ASP1941"[TW] OR "ASP-1941"[TW] OR "ASP 1941"[TW]
	#14	luseogliflozin[TW] OR "1,5-anhydro-1-(5-(4-ethoxybenzyl)-2-methoxy-4-methylphenyl)-1-thioglucitol" [Supplementary Concept] OR "TS-071"[TW] OR "TS 071"[TW] OR "TS071"[TW]
	#15	tofogliflozin[TW] OR "6-((4-ethylphenyl)methyl)-3',4',5',6'-tetrahydro-6'-(hydroxymethyl)spiro(isobenzofuran-1(3H),2'-(2H)pyran)-3',4',5'-triol" [Supplementary Concept] OR "CSG452"[TW] OR "RG7201"[TW]
	#16	ertugliflozin[TW] OR "PF-04971729"[TW] OR "PF 04971729"[TW] OR "LX-4211"[TW]
	#17	"sergliflozin" [Supplementary Concept] OR "sergliflozin"[TW]
	#18	"remogliflozin etabonate" [Supplementary Concept] OR Remogliflozin[TW]
	#19	"(2S,3R,4R,5S,6R)-2-(4-chloro-3-(4-ethoxybenzyl)phenyl)-6-(methylthio)tetrahydro-2H-pyran-3,4,5-triol" [Supplementary Concept] OR Sotagliflozin[TW] OR "LX4211"[TW]
	#20	"Canagliflozin"[Mesh] OR Canagliflozin[TW] OR Invokana[TW]
	#21	#10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20
	#22	#4 AND #9 AND #21
	#23	#22 AND English[Lang]
	#24	#23 NOT (animals[Mesh:noexp] NOT (animals[Mesh:noexp] AND humans[Mesh]))
	#25	#24 AND (Randomized Controlled Trial[ptyp] OR Randomized[TI] OR Randomised[TI] OR Randomly[TI] OR RCT[TI] OR Placebo[TI] OR ((Double*[TI] OR single*[TI] OR treb*[TI] OR tripl*[TI]) AND (Blind*[TI] OR mask*[TI])) OR random*[TI])
Embase	#1	(nondiabet* OR non-diabet* OR without-diabet*):ab,ti,kw
	#2	normal* AND ('glucose'/exp OR glucose*)
	#3	(Prediabet* OR Pre-diabet*):ab,ti,kw
	#4	#1 OR #2 OR #3
	#5	'obesity'/exp
	#6	'overnutrition'/de
	#7	(Obesit* OR Obese OR Overweight* OR 'Over-weight' OR 'Over-weights' OR Overnutrition*):ab,ti,kw
	#8	#5 OR #6 OR #7
	#9	'sodium glucose cotransporter 2'/exp OR ('Sodium Glucose' OR 'Sodium-Glucose' OR 'SGLT2' OR gliflozin):ab,ti,kw
	#10	'dapagliflozin'/exp OR '2 [4 chloro 3 [4 (tetrahydro 3 furyloxy)benzyl]phenyl] 6 hydroxymethyl 2 methoxytetrahydro 2h pyran 3,4,5 triol'/exp OR (dapagliflozin OR 'BMS-512148' OR 'BMS 512148' OR 'BMS512148'):ab,ti,kw
	#11	'empagliflozin'/exp OR ("empagliflozin" OR 'BI-10773' OR 'BI 10773' OR 'BI10773'):ab,ti,kw
	#12	'ipragliflozin'/exp OR ("ipragliflozin" OR "Suglat" OR 'ASP1941' OR 'ASP-1941' OR 'ASP 1941'):ab,ti,kw
	#13	'luseogliflozin'/exp OR (luseogliflozin OR 'TS-071' OR 'TS 071' OR 'TS071'):ab,ti,kw
	#14	'tofogliflozin'/exp OR (tofogliflozin OR 'CSG452' OR 'RG7201'):ab,ti,kw
	#15	'ertugliflozin'/exp OR (ertugliflozin OR 'PF-04971729' OR 'PF 04971729' OR 'LX-4211'):ab,ti,kw
	#16	'sergliflozin etabonate'/exp OR (sergliflozin OR Glucosides):ab,ti,kw
	#17	'remogliflozin etabonate'/exp OR remogliflozin:ab,ti,kw
	#18	'sotagliflozin'/exp OR (Sotagliflozin OR 'LX4211'):ab,ti,kw

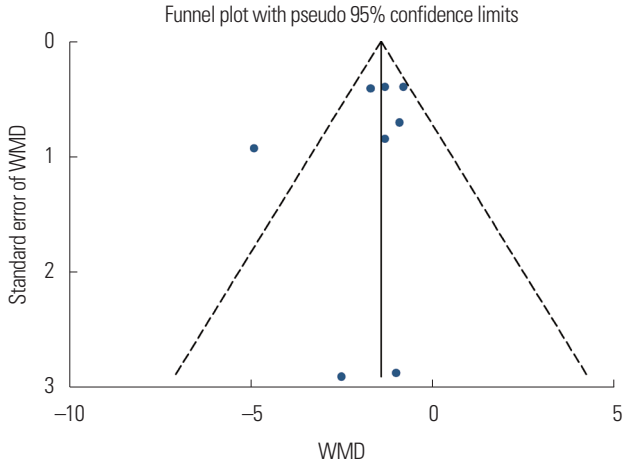
(*Continued to the next page*)

Supplementary Table 1. Continued

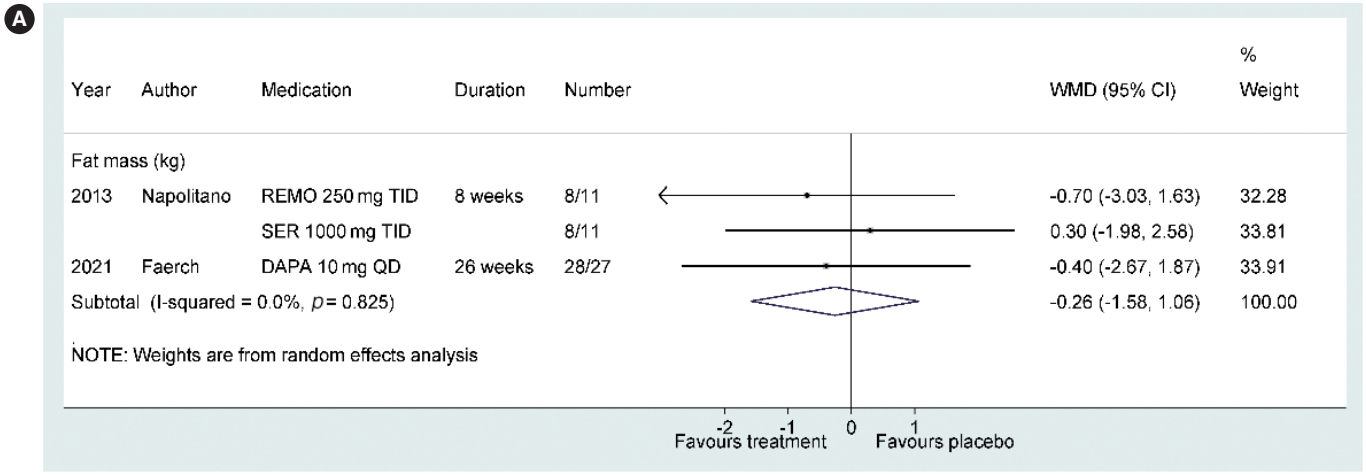
Data source		Search term
	#19	'canagliflozin'/exp OR (Canagliflozin OR Invokana):ab,ti,kw
	#20	#9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19
	#21	#4 AND #8 AND #20
	#22	#21 AND [english]/lim
	#23	#22 NOT ('animal'/de NOT ('animal'/de AND 'human'/exp))
	#24	#23 AND ([article]/lim OR [article in press]/lim OR [review]/lim)
	#25	#24 AND ([randomized controlled trial]/lim OR 'randomized controlled trial'/exp OR 'double blind procedure'/exp OR 'single blind procedure'/exp OR 'triple blind procedure'/exp OR randomized:ti OR randomised:ti OR random*:ti OR placebo:ti)
Cochrane Trias	#1	nondiabet*:ab,ti,kw or non-diabet*:ab,ti,kw or without-diabet*:ab,ti,kw
	#2	normal* and ([mh "glucose"] or glucose*)
	#3	[mh "Prediabetic State"] or Prediabet*:ab,ti,kw or Pre-diabet*:ab,ti,kw
	#4	{OR #1-#3}
	#5	[mh obesity]
	#6	[mh ^overweight]
	#7	[mh ^overnutrition]
	#8	Obesity:ab,ti,kw or Obese:ab,ti,kw or overweight*:ab,ti,kw or over-weight*:ab,ti,kw or "overnutrition":ab,ti,kw or "over-nutrition":ab,ti,kw
	#9	{OR #5-#8}
	#10	[mh "Sodium-Glucose Transporter 2"]
	#11	Sodium Glucose:ab,ti,kw or "SGLT2":ab,ti,kw or gliflozin:ab,ti,kw
	#12	dapagliflozin:ab,ti,kw or "BMS-512148":ab,ti,kw or "BMS 512148":ab,ti,kw or "BMS512148":ab,ti,kw
	#13	empagliflozin:ab,ti,kw or "BI-10773":ab,ti,kw or "BI 10773":ab,ti,kw or "BI10773":ab,ti,kw
	#14	ipragliflozin:ab,ti,kw or "Suglat":ab,ti,kw or "ASP1941":ab,ti,kw or "ASP-1941":ab,ti,kw or "ASP 1941":ab,ti,kw
	#15	luseogliflozin:ab,ti,kw or "TS-071":ab,ti,kw or "TS 071":ab,ti,kw or "TS071":ab,ti,kw
	#16	tofogliflozin:ab,ti,kw or "CSG452":ab,ti,kw or "RG7201":ab,ti,kw
	#17	ertugliflozin:ab,ti,kw or "PF-04971729":ab,ti,kw or "PF 04971729":ab,ti,kw or "LX-4211":ab,ti,kw
	#18	sergliflozin:ab,ti,kw or Remogliflozin:ab,ti,kw or Sotagliflozin:ab,ti,kw or "LX4211":ab,ti,kw
	#19	[mh "Canagliflozin"] or Canagliflozin:ab,ti,kw or Invokana:ab,ti,kw
	#20	{OR #10-#19}
	#21	#4 and #9 and #20
	#22	#21 in Trials (exclude ongoing trial, Conference paper)
	#23	#21 and (Random* or RCT or Placebo* or ((Double* or single* or treb* or tripl*) near/6 (Blind* or mask*)))
	#24	#23 in Trials (exclude ongoing trial, Conference paper)
Web of Science	#1	TS=(nondiabet* OR non-diabet* OR without-diabet* OR (normal* AND glucose*) OR Prediabet* OR Pre-diabet*)
	#2	TS=(Obesit* OR Obese OR Overweight* OR Overnutrition* OR over-weight*)
	#3	TS=("Sodium Glucose" OR "SGLT2" OR gliflozin OR dapagliflozin OR "BMS-512148" OR "BMS 512148" OR "BMS512148" OR "empagliflozin" OR "BI-10773" OR "BI 10773" OR "BI10773" OR "ipragliflozin" OR "Suglat" OR "ASP1941" OR "ASP-1941" OR "ASP 1941" OR luseogliflozin OR "TS-071" OR "TS 071" OR "TS071" OR tofogliflozin OR "CSG452" OR "RG7201" OR ertugliflozin OR "PF-04971729" OR "PF 04971729" OR "LX-4211" OR "sergliflozin" OR Remogliflozin OR Sotagliflozin OR "LX4211" OR Canagliflozin OR Invokana)
	#4	#1 AND #2 AND #3
	#5	#4 Refined by: DOCUMENT TYPES: (ARTICLE OR EARLY ACCESS OR REVIEW)
	#6	#5 AND TS=(random* OR placebo* OR RCT)

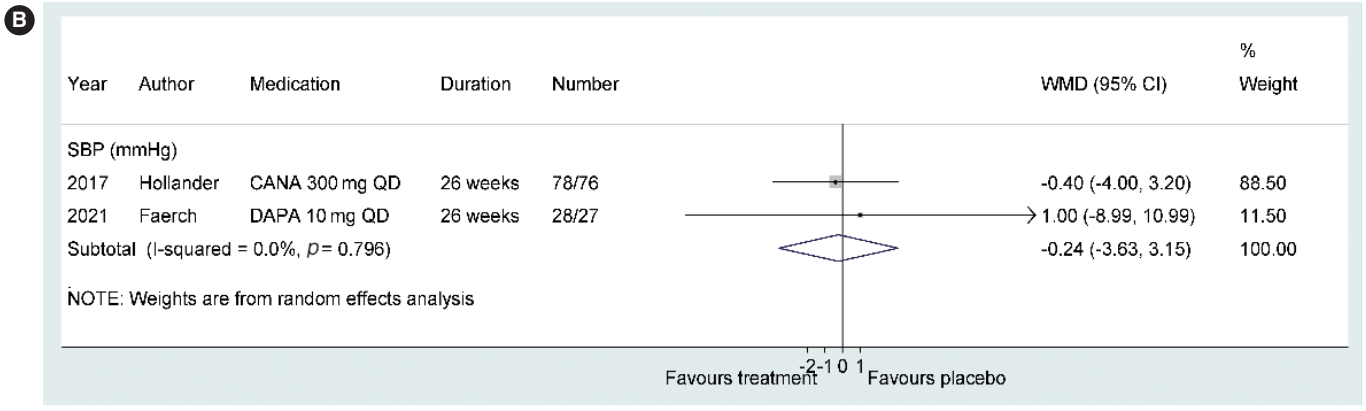


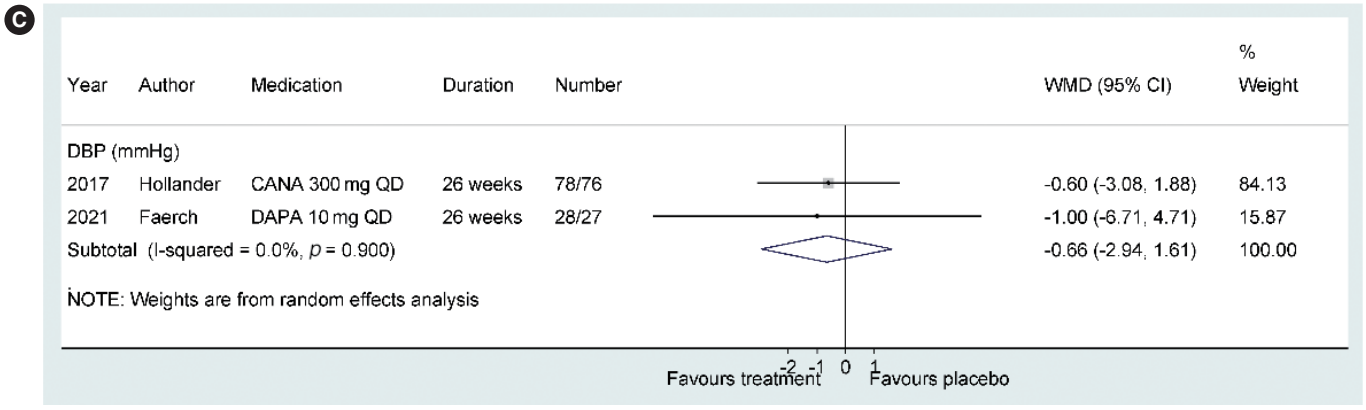
Supplementary Figure 1. Risk of bias in the included studies. NA, not available.



Supplementary Figure 2. Funnel plot for body weight change in the included studies. WMD, weighted mean difference.

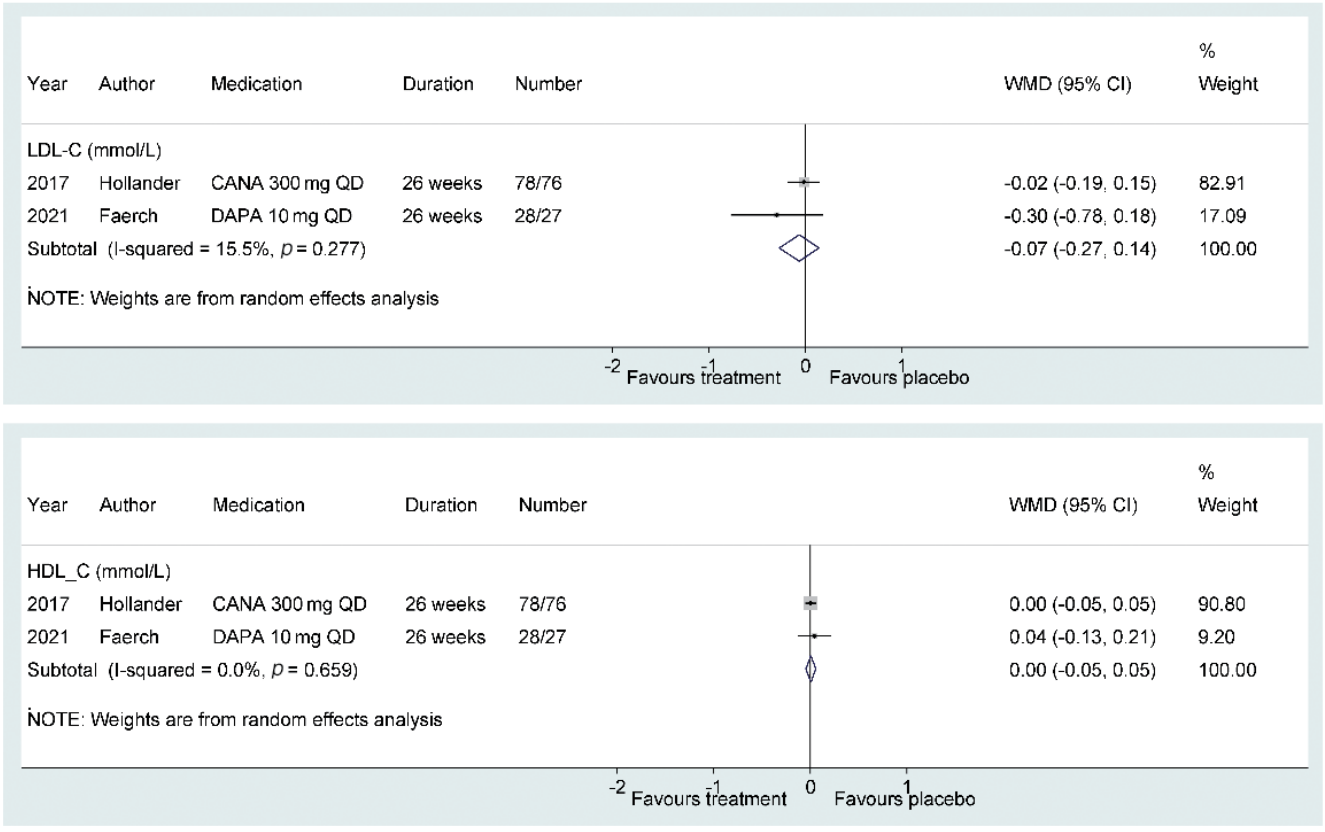






Supplementary Figure 3. Forest plots of sodium-glucose cotransporter 2 inhibitor-induced change in (A) fat mass, (B) systolic blood pressure (SBP), (C) diastolic blood pressure (DBP).

(*Continued to the next page*)



Supplementary Figure 3. (Continued) (D) low-density lipoprotein cholesterol (LDL-C), and (E) high-density lipoprotein cholesterol (HDL-C) compared with placebo. WMD, weighted mean difference; CI, confidence interval; TID, three times a day; QD, once a day.